SCI-FORGE 2026 · STUDENT SCIENCE EXHIBITION

# Cosmos

One continuous flight from the International Space Station to the oldest light in the universe, built from real data and running live in a browser.




cosmos-astronexus.vercel.app

# Space covers too many scales to picture.

**109 m**

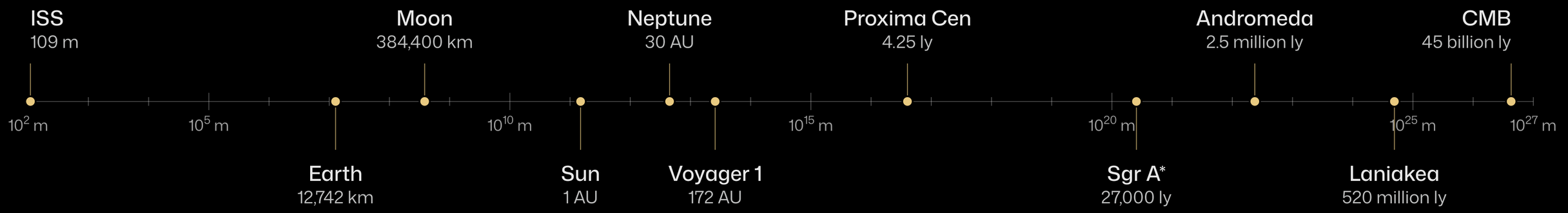
the International Space Station, end to end

**45 billion ly**

to the shell of the cosmic microwave background

**≈ 25**

powers of ten between them. A printed diagram can show about two before things vanish or fall off the page.




Sizes and distances: NASA fact sheets, IAU 2015, Gaia DR3, Planck 2018, as cited in the app's tour narration.

03 WHAT COSMOS IS

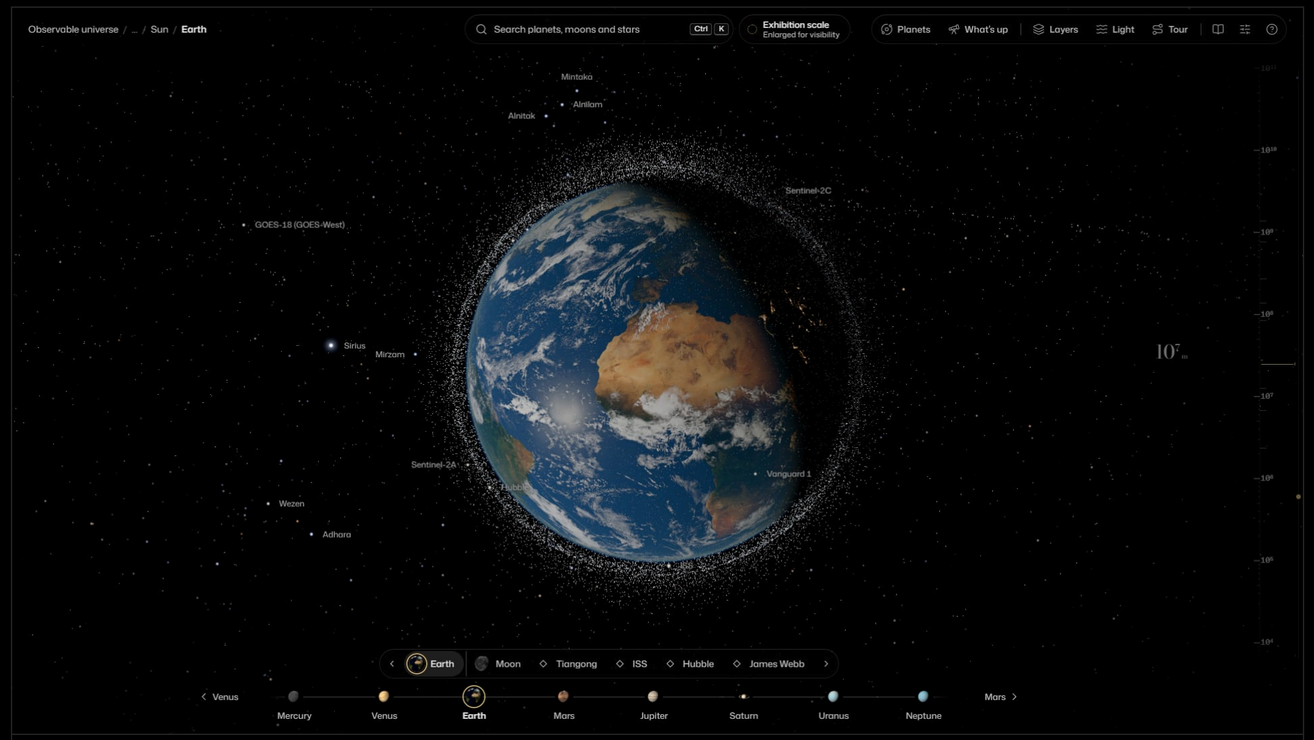
# A 3D universe explorer with one continuous camera, from Earth orbit to the cosmic microwave background, that puts every object where real data says it is today.

**Real positions, now.** Planets and moons from ephemeris formulas; about 31,800 satellites and debris propagated with SGP4 from CelesTrak elements.

**Every value sourced.** Click anything: each number is marked observed, derived, reconstructed or illustrative.

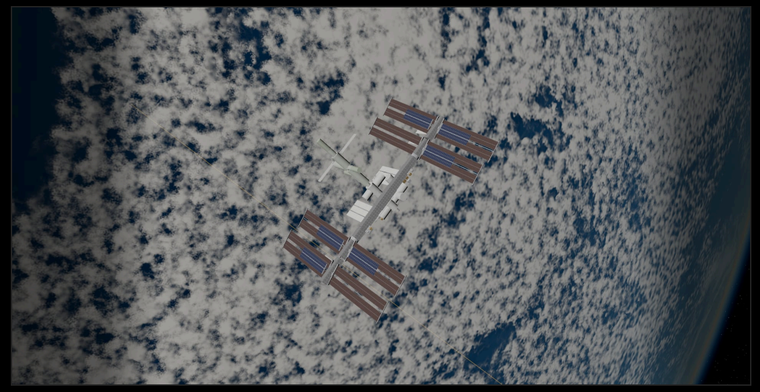
**Four labs.** Orbital debris, asteroid deflection, habitability, black holes.

**Offline in a browser.** 76 datasets on disk, no account, no server.




The live app at Earth: search at the top, the Exhibition scale · Enlarged for visibility disclosure beside it, the planet bar and the moons and stations of the body in focus at the bottom, the Cosmic Ruler on the right.

# From home to the edge of the Sun's wind



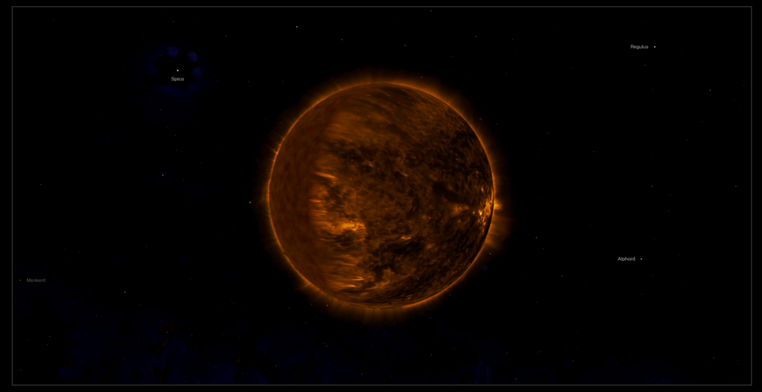
## International Space Station

about 400 km up · one orbit every 92 min



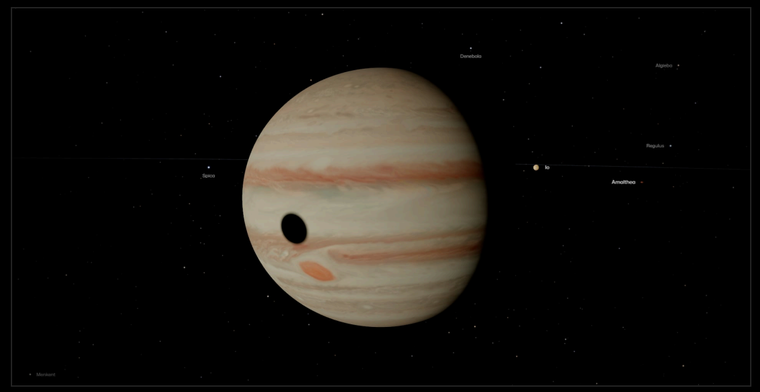

## The Moon

384,400 km · 1.3 light-seconds



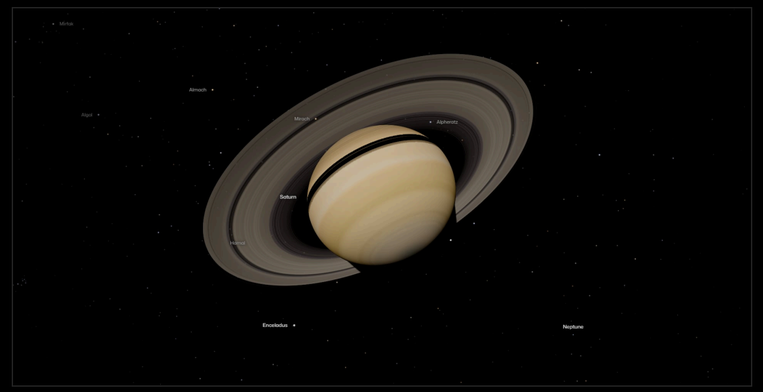

## The Sun

1 AU = 149.6 million km · 1.39 million km wide



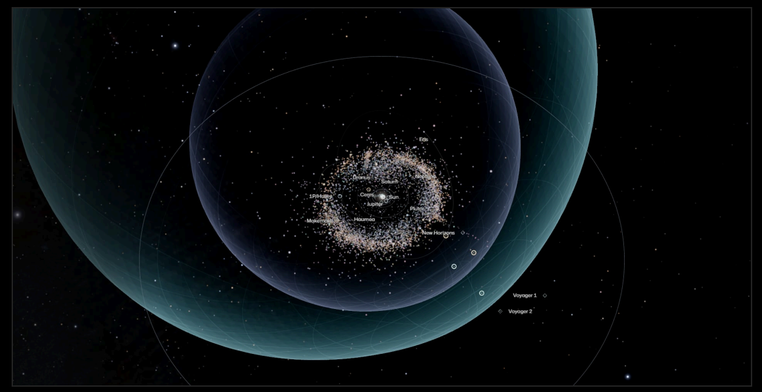

## Jupiter

5.20 AU · 11.2 Earths wide




## Saturn

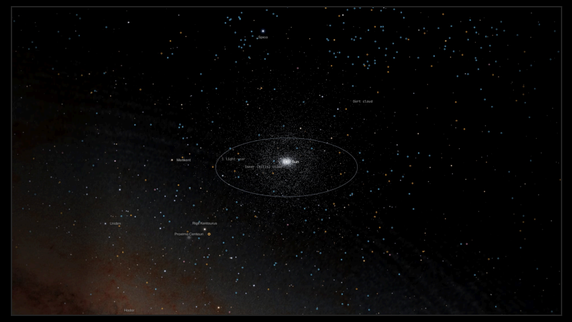
9.54 AU · rings with the Cassini division




## Looking back from Voyager 1

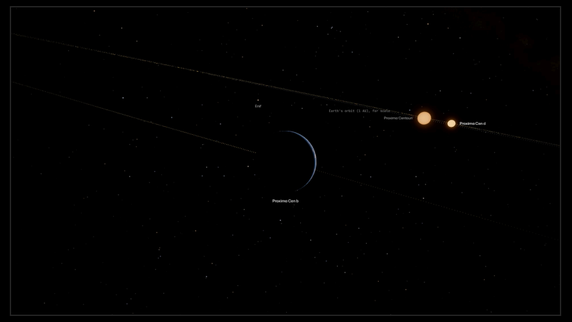
172 AU · radio takes 23.9 hours

# Past the stars to the oldest light



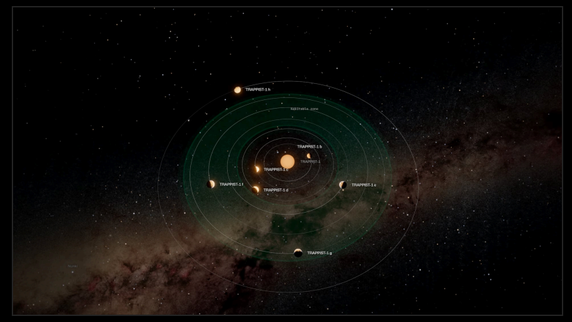
## Oort cloud

up to ~1.6 ly · illustrative



## Proxima Centauri

4.25 ly · Voyager 1: 75,000 years



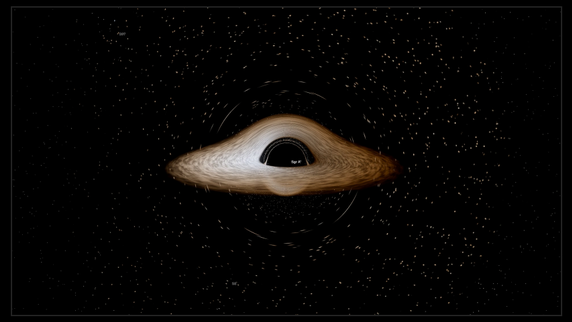
## TRAPPIST-1

40.7 ly · seven rocky planets



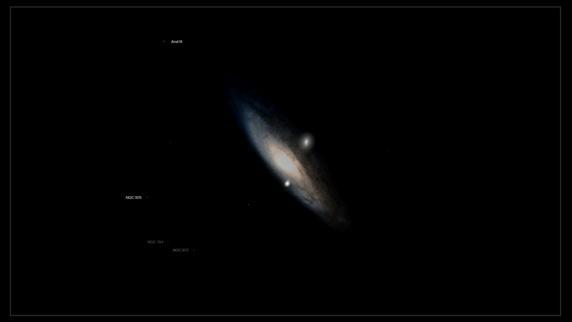
## The Milky Way

Sun 27,000 ly out · reconstructed



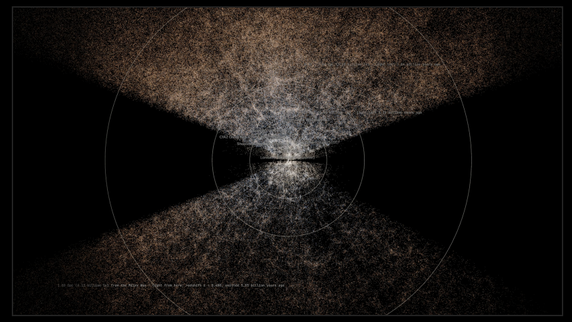
## Sagittarius A*

4.3 million Suns · horizon 25 million km



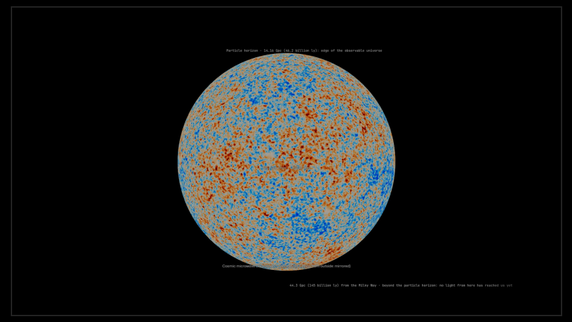
## Andromeda

2.5 million ly · about a trillion stars



## The cosmic web

2.2 million real galaxies · 6.5 billion ly



## Cosmic microwave background

45 billion ly · light 13.8 billion years old

23 chapters · about nine minutes · press T in the app

06 SCIENCE AND PROVENANCE

# Real catalogues, and an honest label on every number

**31,800**
tracked satellites and debris

**75,855**
asteroids, plus 3,449 near-Earth objects

**109,389**
stars (HYG v4.4, Gaia)

**6,451**
exoplanets around 4,847 stars

**2.2 million**
galaxies of the cosmic web (2MRS, DESI DR1)

**76**
datasets, each with date, licence and citation

## EVERY FACT CARRIES ONE OF FOUR MARKS

| | |
|---|---|
| **Observed** | a measured value from a cited source |
| **Derived** | computed from observed values with a stated model |
| **Reconstructed** | inferred from data, like the Milky Way seen from outside |
| **Illustrative** | drawn for clarity: Oort cloud, comet tails, exoplanet surfaces, black-hole discs |

A missing value reads **Unknown**, never 0.

**Enlarged for visibility**

Bodies are drawn larger by default (Earth ≈ 448× at 4 AU) and shrink to true size within 40 radii. The label is always on screen; one click shows true scale. Distances are never changed.

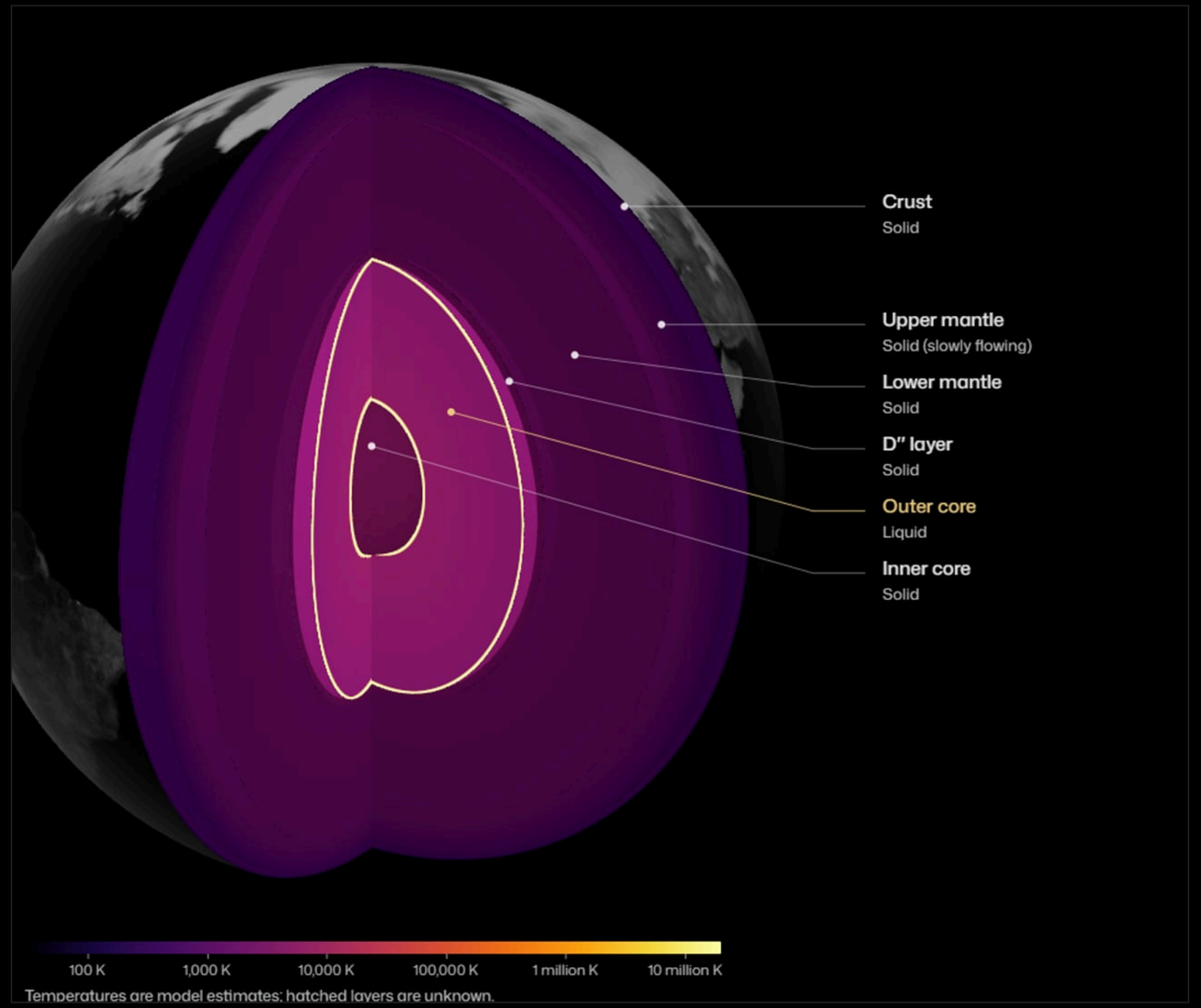
07 DEEP DIVE · INSIDE THE EARTH

# Mapped by earthquake waves

The cutaway uses PREM, the Preliminary Reference Earth Model. Shear waves cannot cross the outer core, which is how we know it is liquid.

| LAYER | OUTER RADIUS | STATE |
|---|---|---|
| Crust | 6,371 km | solid |
| Upper mantle | 6,346.6 km | solid, slowly flowing |
| Lower mantle | 5,701 km | solid |
| D″ layer | 3,630 km | solid |
| Outer core | 3,480 km | liquid iron, 3,800–6,200 K |
| Inner core | 1,221.5 km | solid, 12,760–13,090 kg/m³ |

18 bodies · 67 layers · 50 references. Layers thinner than 1.4% of the radius are drawn thicker, and the view says so.



# A star that weighs a black hole

The star S2 circles something invisible every 16 years. Only a black hole of 4.3 million Suns fits in the space.

**16.05 yr**
orbital period

**120 AU**
closest approach

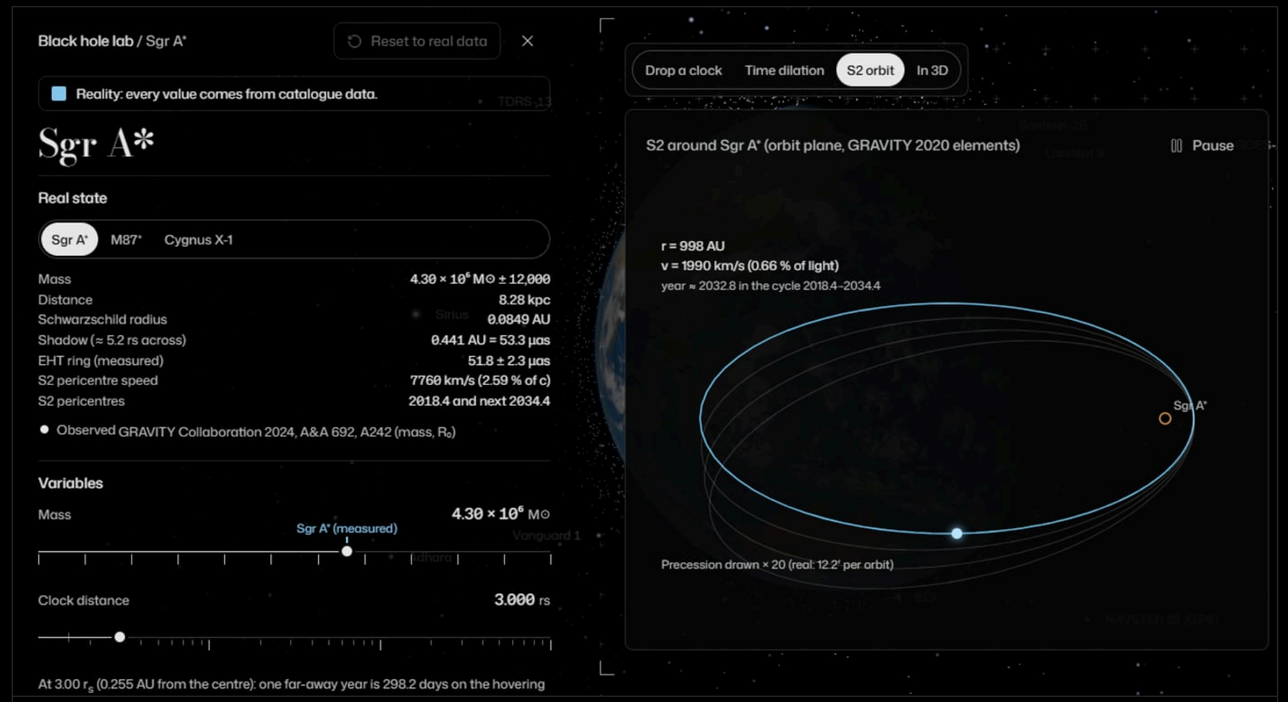
**7,700 km/s**
2.6% of the speed of light

**12′**
orbit precession per loop (relativity)

**≈ 200 km/s**
redshift at pericentre

**2034**
next pericentre

GRAVITY Collaboration 2018, 2020, 2024. Discovery: Nobel Prize in Physics 2020 (Genzel, Ghez).




The black hole lab: measured values for Sgr A* on the left, S2 on its GRAVITY 2020 orbit, with the precession drawn 20 times larger so it can be seen.

09 DEEP DIVE · BLACK HOLES

# Traced, not painted

The shader marches light rays past the hole and bends the real captured sky behind it.

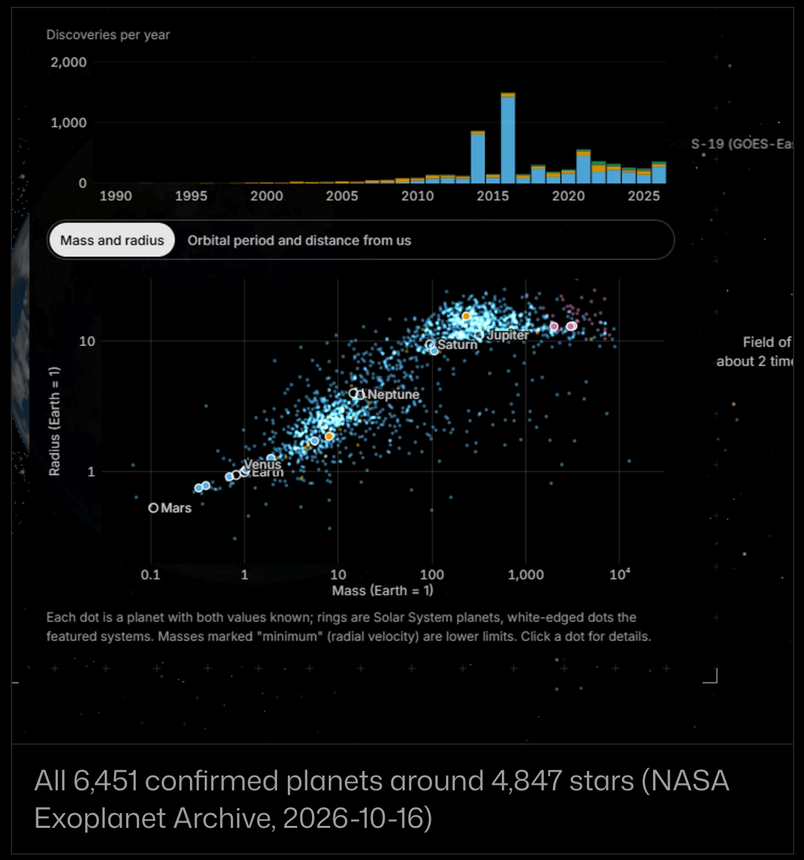
- Shadow, photon ring, secondary images, Einstein rings
- Approximate Kerr bending when the spin is measured
- Disc colour from blackbody, Doppler beaming, gravitational redshift
- 13 measured black holes, from Gaia BH1 to M87*

The bending is computed; the disc's look is illustrative. Cost: about 0–4 ms per frame on Balanced.

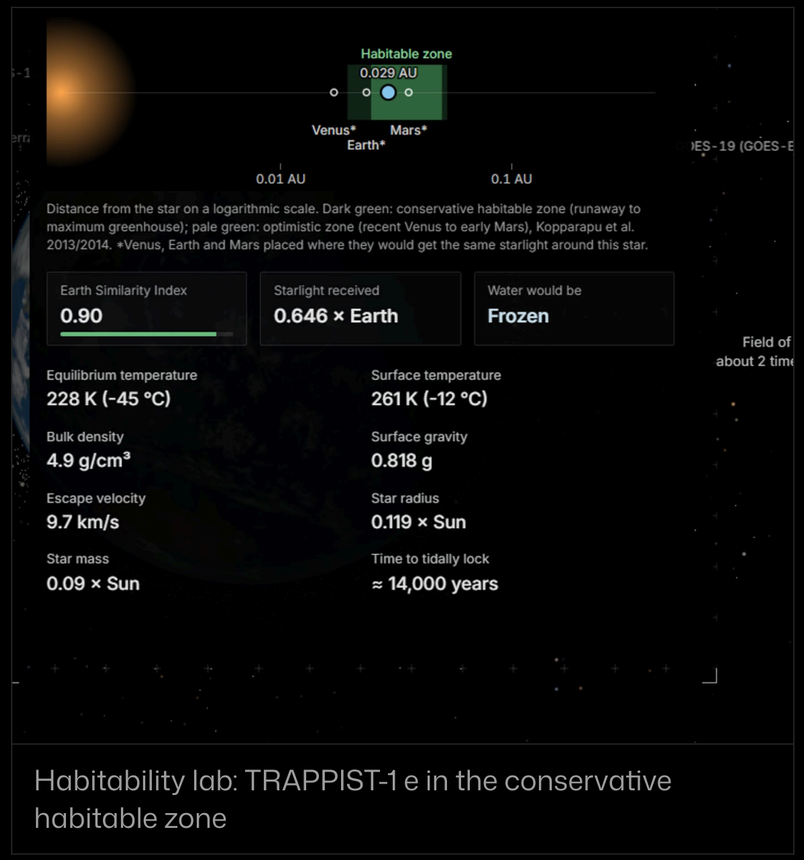



Gaia BH1: an Einstein ring of background stars

# Could it keep liquid water?




All 6,451 confirmed planets around 4,847 stars (NASA Exoplanet Archive, 2026-10-16)




Habitability lab: TRAPPIST-1 e in the conservative habitable zone

**255 K → 288 K**

the lab's Earth, without and with its atmosphere

**0.95–1.67 AU**

the Sun's conservative habitable zone (Kopparapu et al. 2014)

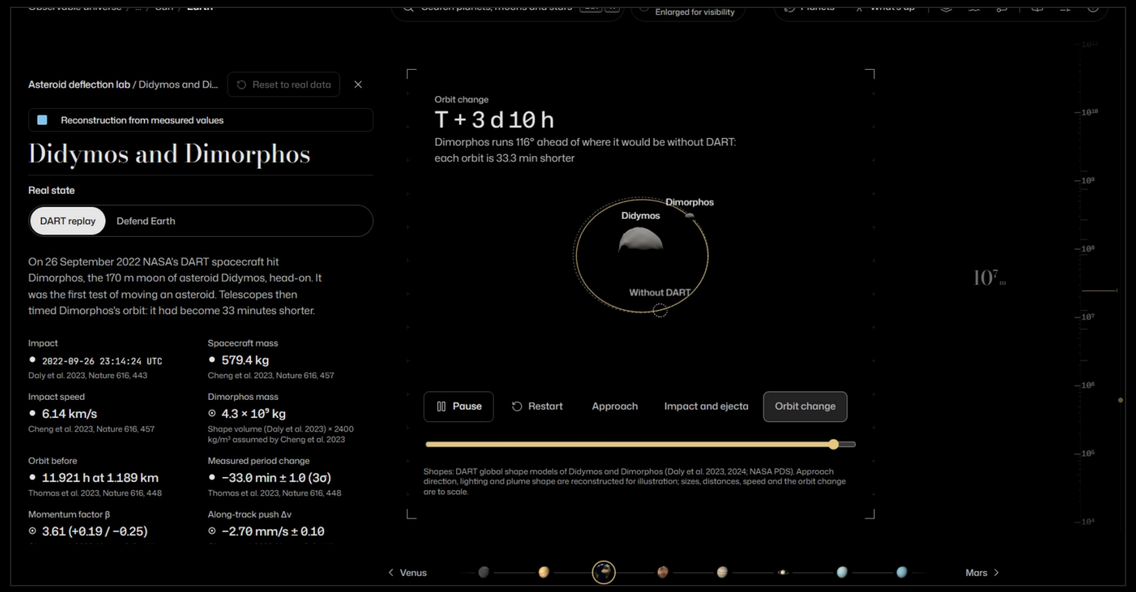
**14**

showcase systems drawn on their archive orbits

Planet surfaces are an artist's impression: no exoplanet has been imaged as a disc.

# Change a variable, watch the result




## Orbital debris

Starts from the real catalogue: 29,408 tracked objects below 2,000 km. Then launches, drag, removal and collisions run forward 100 years.

**≈ 583,000**
objects by 2126, business as usual

**≈ 25,000**
if launches stop

## Asteroid deflection

Replays NASA's DART impact on Dimorphos (26 September 2022), then lets you defend Earth from a fictional asteroid.

**−33.3 min**
our model's orbit change

**−33.0 ± 1.0**
measured (Thomas et al. 2023)

Also: the habitability lab (slide 10) and the black hole lab (slide 8). All are teaching models, and they say so.

# 25 powers of ten at 60 frames per second

## Continuous scale

*render = (world − camera) ÷ metres per unit*

Positions stay in 64-bit on the CPU; the camera sits at the origin and the scale runs from $10^{-3}$ to $10^{30}$ m per unit. A logarithmic depth buffer keeps near and far in order.

## Texture streaming

1,155 → 603 MB

KTX2 GPU-compressed textures under a per-preset VRAM budget, 8K for the body in focus. Frames over 100 ms fell from 33 to 13 on a nine-body tour.

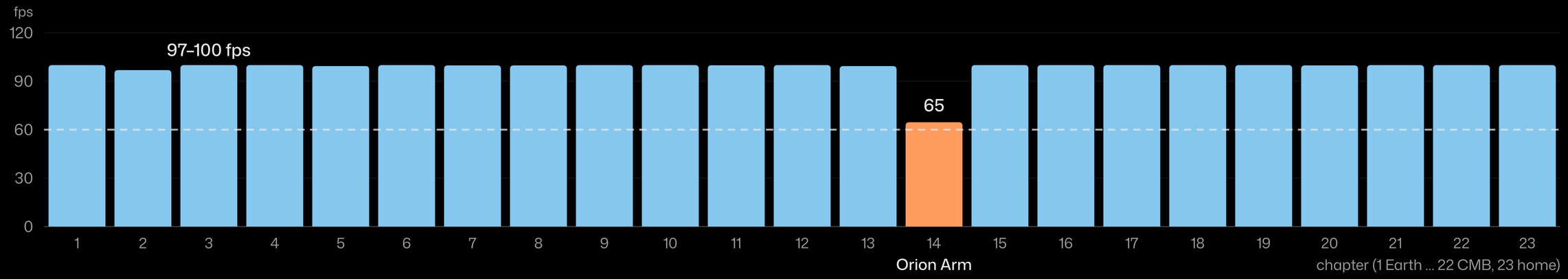
## Off the main thread

643 ms → 0

Shaders link in the background before first use. SGP4 for 31,800 objects, the 76,000-row asteroid table and Andromeda's star cloud are built in web workers.

### Frame rate in every chapter of the golden journey

Production build, 1920×1080, Ultra, headless Chrome on the stand's RTX 3050 laptop. Dashed line: 60 fps target.




Vite · React 19 · TypeScript strict · three.js · about 72,000 lines · 635 unit tests checked against published values

# Built for a crowded exhibition floor

| | |
|---|---|
| **Kiosk start** | `?exhibition=1` boots with larger type and touch targets, read from 1–2 m on a 1920×1080 touchscreen. |
| **Reduce motion** | Camera flights become a short crossfade. |
| **Adapts to the hardware** | Auto quality lowers detail to hold 60 fps; Fast, Balanced and Ultra presets. |
| **Works with no internet** | No request leaves the laptop; an end-to-end test blocks every outside request while it visits every region, lab and tour. |
| **Reset between visitors** | H flies home to Earth, Esc closes everything, Shift + H hides the interface. |

## EVERYTHING HAS A KEY

| | |
|---|---|
| T | golden journey tour |
| ← → | previous / next chapter |
| 1 – 8 | fly to a planet |
| [ ] | step through its moons |
| Ctrl K | search anything |
| B | navigator |
| N | time to now |
| ? | help |

# Five things this project taught us

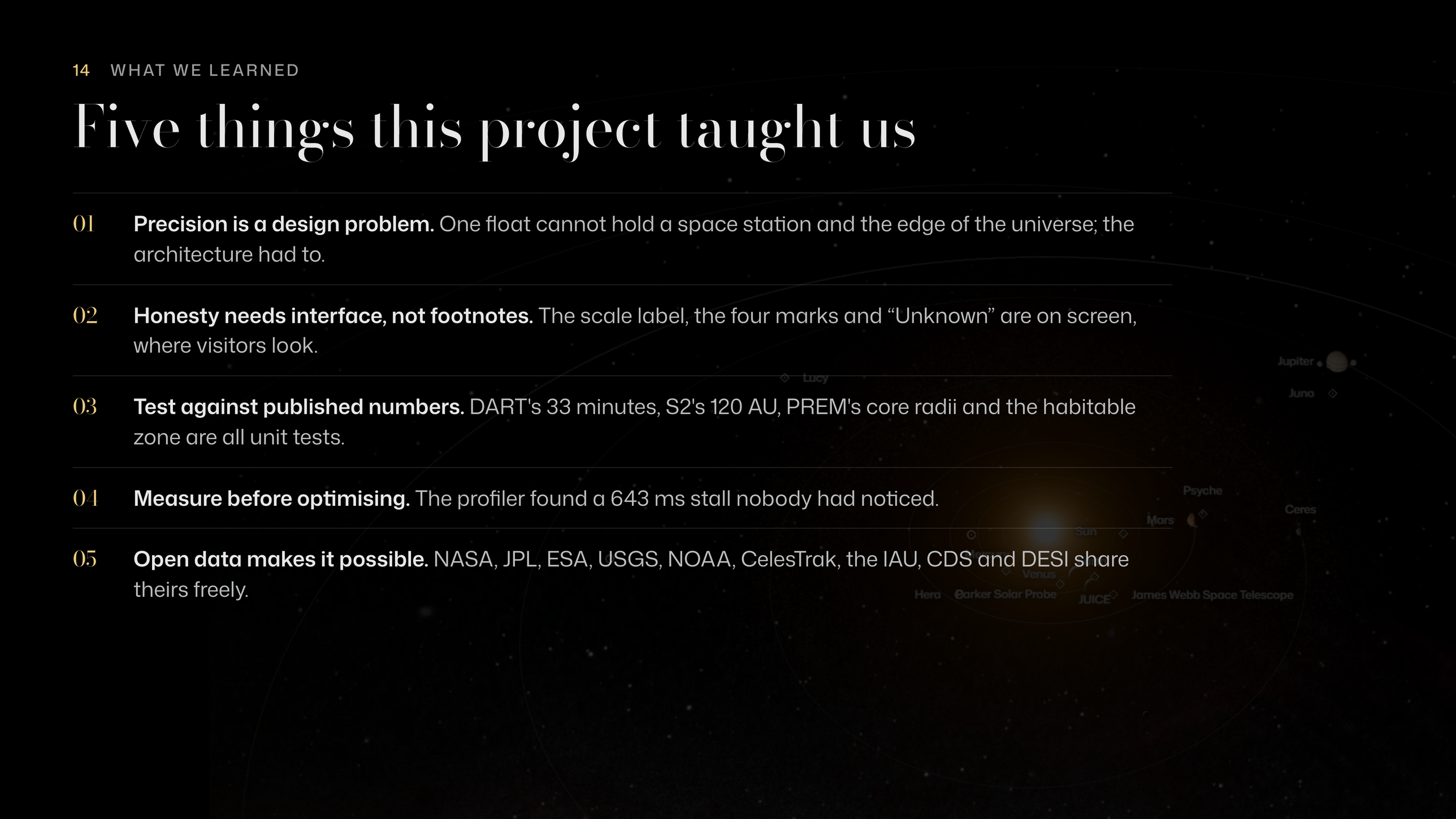
01 **Precision is a design problem.** One float cannot hold a space station and the edge of the universe; the architecture had to.

02 **Honesty needs interface, not footnotes.** The scale label, the four marks and “Unknown” are on screen, where visitors look.

03 **Test against published numbers.** DART's 33 minutes, S2's 120 AU, PREM's core radii and the habitable zone are all unit tests.

04 **Measure before optimising.** The profiler found a 643 ms stall nobody had noticed.

05 **Open data makes it possible.** NASA, JPL, ESA, USGS, NOAA, CelesTrak, the IAU, CDS and DESI share theirs freely.



SCI-FORGE 2026

# Thank you

Fly it yourself, or press T at our stand.

---

cosmos-astronexus.vercel.app

Scan to open Cosmos

